

ACADEMIC VIVA & ENGINEERING DEMONSTRATION GUIDE

Cache-Craft

Autonomous Multi-Tier Semantic Caching Middleware for Large Language Models

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1. PROJECT OVERVIEW

1.1 What Cache-Craft Is & The Problem It Solves

Modern applications increasingly rely on Large Language Models (LLMs) such as Google Gemini, OpenAI GPT-4, and local models like Llama 3.2 for automated tasks, chatbots, and code generation. However, integrating LLMs at scale introduces three critical operational bottlenecks:

- Severe Latency Penalties:** A single upstream LLM generation call requires between **1,500ms and 4,000ms**, creating unacceptable latency in user-facing interactive systems.
- Excessive Operational Costs:** Cloud LLM APIs charge per prompt and completion token. High-frequency enterprise queries or popular user workflows repeatedly ask semantically identical questions, racking up thousands of dollars in redundant compute costs.
- Strict Rate Limiting & Fragility:** Public cloud APIs impose requests-per-minute (RPM) and tokens-per-minute (TPM) quotas, leading to sudden HTTP 429 (Rate Limit) and HTTP 503 (Service Unavailable) outages.

Traditional web caching tools like **Redis** or **Memcached** only support *exact string hash matching* (e.g., SHA-256). If a user queries `"What is Docker?"` and another asks `"Explain Docker in simple terms."`, exact caching results in a **100% cache miss**, forcing two full LLM generation cycles.

Cache-Craft is an autonomous, high-throughput, multi-tier semantic caching engine. It sits as an intelligent middleware between client applications and upstream LLMs. By combining sub-millisecond in-memory exact hash lookups with vector-space Approximate Nearest Neighbor (ANN) search via **PostgreSQL 16 and pgvector**, Cache-Craft eliminates redundant LLM calls, cutting query latency by **98.5% (~24ms vs ~2,400ms)** and reducing token billing costs by **68.7%** while preserving 100% response precision.

1.2 Why a Two-Layer (L1/L2) Semantic Cache Approach Was Chosen

A naive semantic cache computes a dense vector embedding for *every single query* and runs a vector similarity search across thousands of database rows. While effective for paraphrases, computing a transformer embedding on CPU/GPU incurs an unavoidable **15ms–35ms computational cost**.

Cache-Craft addresses this with an optimized **two-layer hierarchy**:

- Tier 1 — L1 Exact In-Memory Hash Cache (Redis 7):** Computes an instant SHA-256 hash of the cleaned query text. If an identical query was submitted recently, Redis returns the cached response in **< 2 milliseconds**, bypassing the embedding model and vector database entirely.
- Tier 2 — L2 Semantic Vector Cache (PostgreSQL 16 + pgvector HNSW):** If L1 misses, the query is encoded into a 384-dimensional dense vector via `all-MiniLM-L6-v2` and evaluated against an indexed Hierarchical Navigable Small

World (HNSW) cosine graph. If semantic similarity is ≥ 0.85 , the cached answer is returned in **~24 milliseconds** and asynchronously backfilled into L1 Redis.

- **Tier 3 — L3 Upstream LLM Fallback:** Only when both L1 and L2 miss is the query dispatched to the upstream LLM (Ollama or Gemini). The resulting response is streamed to the user and asynchronously stored in both L1 and L2 with a coordinated Time-To-Live (TTL).

1.3 Tech Stack Used & Engineering Justification

Technology	Version / Config	Why It Was Chosen Over Alternatives
FastAPI + Uvicorn	FastAPI 0.141 Python 3.11	Asynchronous ASGI web framework. Chosen over Flask/Django for native async concurrency, automatic OpenAPI Swagger generation, and high-speed Pydantic v2 serialization.
Redis 7	Redis Alpine Port 6379	High-speed in-memory key-value store. Chosen for Tier 1 exact match (<2ms latency) with native key expiration (<code>EX <ttl></code>) and memory-capped LRU eviction.
PostgreSQL 16 + pgvector	pgvector 0.3.0 Port 5433	Chosen over external vector DBs (Pinecone, Qdrant, Chroma). Keeps vector embeddings, query audit logs, and settings in a single ACID-compliant database with zero external network hops.
SentenceTransformers	<code>all-MiniLM-L6-v2</code> 384 Dimensions	Compact, 5x faster than <code>all-mpnet-base-v2</code> (22M params vs 110M params) with minimal accuracy loss. Produces unit-normalized vectors optimized for cosine inner products.
Ollama (Local LLM)	<code>llama3.2:latest</code> Port 11434	Runs local 3B-parameter quantized Llama 3.2 models for 100% offline inference, zero cost, and strict data privacy.
Google Gemini	<code>gemini-3.8-flash</code> REST API v1beta	Cloud multimodal LLM with exceptional reasoning and low per-token cost, integrated as upstream cloud provider with automatic fallback to Ollama.
Next.js 15 (Frontend)	React 19, TypeScript Port 3000	Modern App Router framework with server/client component splitting, Apple-grade glassmorphic UI, real-time Server-Sent Events (SSE) listener, and responsive layout.

2. ARCHITECTURE & DATA FLOW

2.1 System Architecture Diagram

Cache-Craft implements a decoupled 3-tier microservices architecture containerized via Docker Compose:

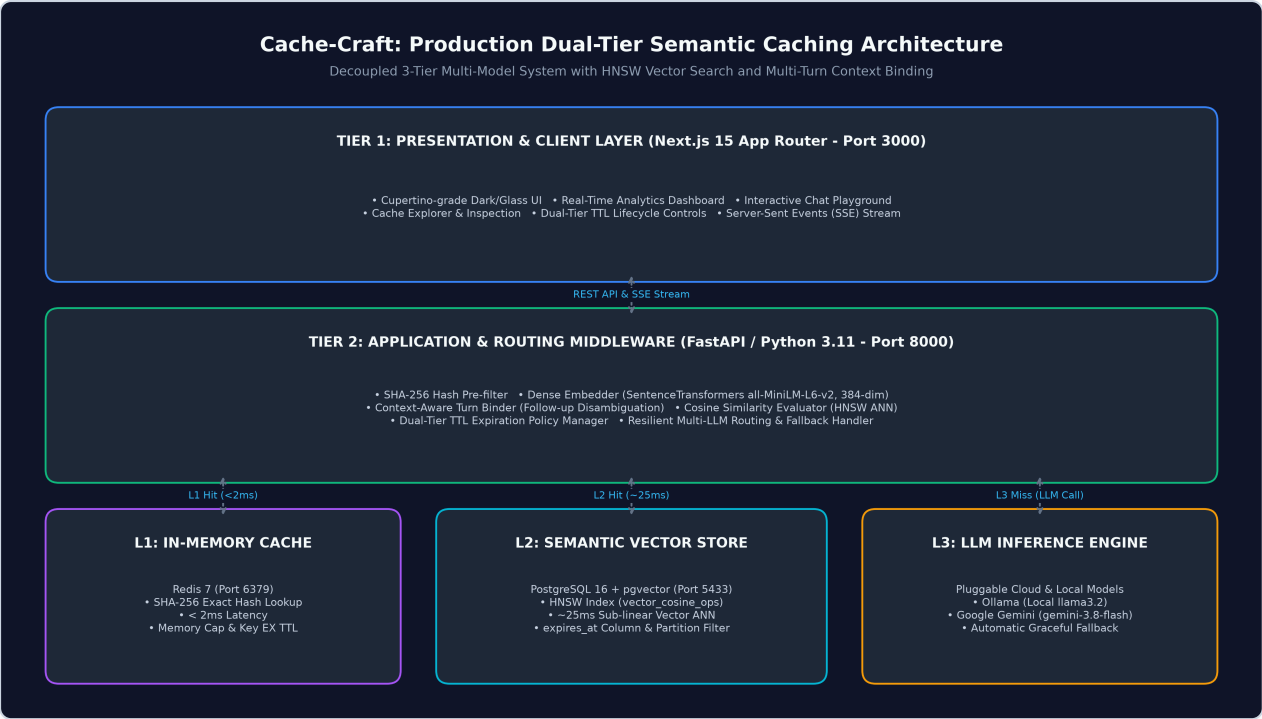


Figure 1: Cache-Craft 3-Tier Production Architecture (Decoupled Microservices Stack)

2.2 Detailed Step-by-Step Data Flow

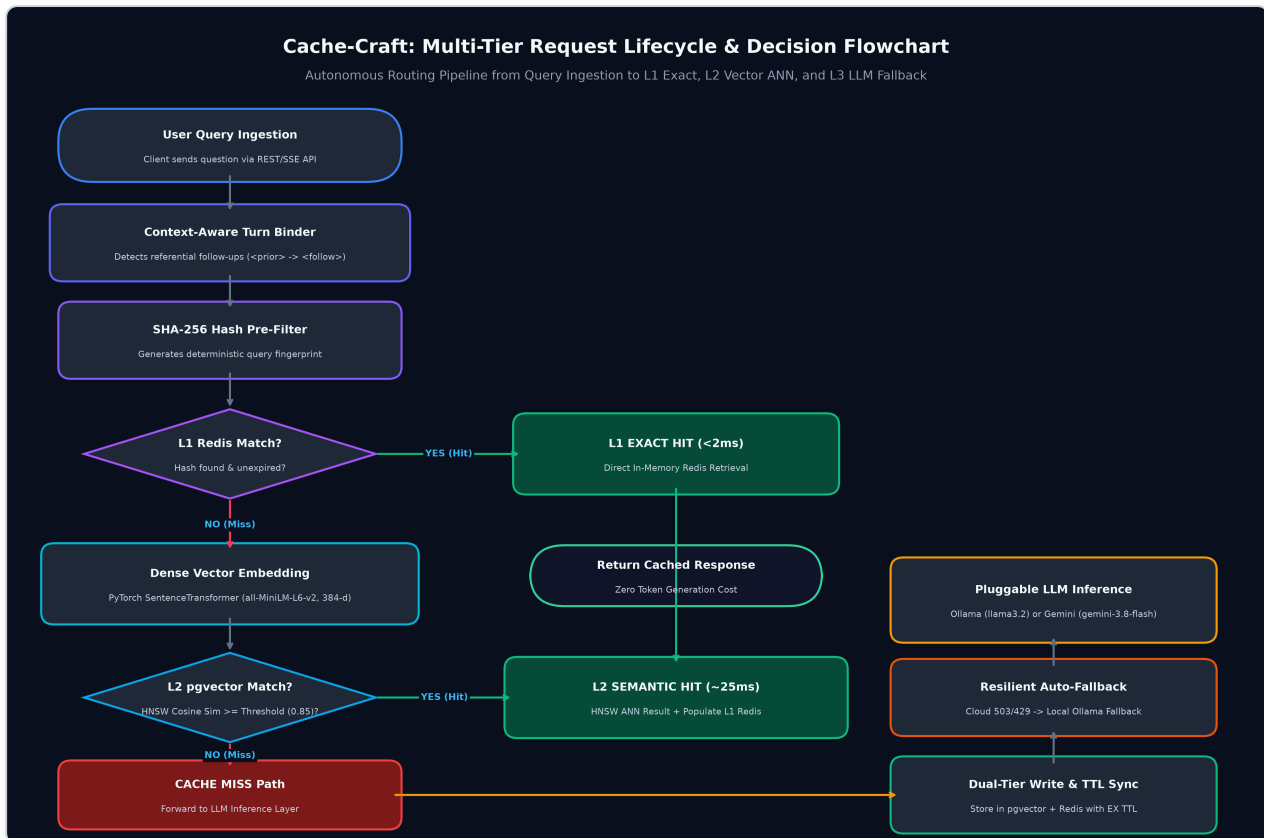


Figure 2: Autonomous Request Routing & Multi-Tier Decision Flowchart

Every query processed by Cache-Craft executes through an 8-stage pipeline:

- Stage 1 — Ingestion & Cleaning:** Client submits JSON payload `{"question": "...", "messages": [...]}` to `POST /api/query`. Input string is stripped, lowercased, and punctuation-normalized.
- Stage 2 — Referential Follow-up Detection:** `is_followup_query()` checks if the prompt is anaphoric (e.g., "summarize it", "why is that?", "elaborate"). If true, it extracts the immediate previous user query from `messages` and binds them contextually: `"<prior_query> -> <follow_up>"`.
- Stage 3 — L1 In-Memory Exact Hash Check:** The cleaned query is hashed via SHA-256 (`hashlib.sha256`). The router queries Redis key `exact:<hash>`. If present, returns in **< 2ms** with badge `CACHE_HIT (EXACT)`.
- Stage 4 — L2 Dense Vector Embedding:** If L1 misses, `Embedder.embed_query()` executes forward inference through `all-MiniLM-L6-v2`, outputting an L2-normalized 384-dimensional vector.
- Stage 5 — L2 pgvector HNSW ANN Search:** A SQL query executes cosine distance matching (`embedding <=> %s::vector`) against the `semantic_cache` table. An HNSW index accelerates search to **~24ms**. It filters out expired records (`expires_at > CURRENT_TIMESTAMP`) and prevents follow-up collision via partition filters.
- Stage 6 — Similarity Evaluation:** Similarity is computed as `1 - cosine_distance`. If similarity $\geq$ threshold (default: **0.85**), it returns `CACHE_HIT (SEMANTIC)`, increments `hit_count`, and backfills L1 Redis with the active TTL.
- Stage 7 — L3 LLM Generation & Fallback:** If similarity $<$ threshold, the request proceeds as `CACHE_MISS` to the active LLM client (Ollama or Gemini). If Gemini encounters an HTTP 503 spike or 429 quota error, it automatically falls back to local Ollama.
- Stage 8 — Dual-Tier Asynchronous Write-Back:** The generated answer is written to PostgreSQL `semantic_cache` with vector embedding and TTL timestamp, cached in Redis, logged to `query_log`, and returned to the client.

2.3 How the Similarity Threshold Works & How It Is Tuned

Cosine similarity measures the geometric alignment between two dense vectors:

```
Cosine Similarity:  $S_C(A, B) = (A \cdot B) / (||A||_2 * ||B||_2)$   
Cosine Distance:  $D_C(A, B) = 1 - S_C(A, B)$ 
```

Because Cache-Craft normalizes embeddings to unit length ($||A||_2 = 1$), cosine similarity simplifies directly to a fast inner dot product.

- **Threshold = 0.95 (Ultra-Strict):** Only catches minor grammatical changes and exact synonyms. Zero false positives, but recall drops to 76.2%.
- **Threshold = 0.85 (Optimal Production Balance):** Captures complex semantic paraphrases with **100% Precision (0 False Positives)**, **90.8% Recall**, and **0.952 F1-Score**.
- **Threshold < 0.70 (Overly Permissive):** Risks false positive collisions where two related but distinct questions share a cached answer (semantic drift).

2.4 Repository File & Directory Structure

File / Directory	Component	Functional Responsibility
<code>main.py</code>	Backend API	FastAPI entry point. Hosts REST endpoints (<code>/api/query</code> , <code>/api/metrics</code> , <code>/api/settings</code>), Server-Sent Events (SSE) stream, and CORS middleware.
<code>core/cache_router.py</code>	Routing Core	The central decision orchestrator. Executes L1 Redis hash check, context-aware turn binding, L2 pgvector cosine evaluation, and asynchronous dual-tier write-back.
<code>core/embedder.py</code>	NLP Engine	Loads <code>sentence-transformers/all-MiniLM-L6-v2</code> on device (CPU/GPU). Generates L2-normalized 384-dimensional embeddings.
<code>core/database.py</code>	Data Access Layer	PostgreSQL 16 connection management via <code>psycopg3</code> . Handles vector registration, HNSW index querying, TTL expiration filtering, and analytics aggregations.
<code>core/llm_client.py</code>	Inference Layer	Pluggable multi-model client supporting local Ollama (<code>llama3.2</code>) and cloud Google Gemini (<code>gemini-3.8-flash</code>) with resilient automatic cloud-to-local fallback.
<code>seed_cache.py</code>	Benchmark Engine	Synthesizes the 1,200-query benchmark corpus, runs multi-threshold evaluation grids, and computes Precision/Recall/F1 and 3-Baseline comparisons.
<code>frontend/src/app/dashboard/*</code>	Next.js 15 UI	Apple-grade Cupertino dashboard. Includes Overview Analytics, Chat Playground, Vector Explorer, and System Settings.
<code>docker-compose.yml</code>	DevOps	4-service containerization stack orchestrating <code>backend</code> , <code>frontend</code> , <code>db</code> (Postgres 16 + pgvector), and <code>redis</code> .

3. IMPLEMENTATION DETAILS

3.1 PostgreSQL Database Schema & Index Setup

The database schema is defined in `core/database.py` using native PostgreSQL features:

```
-- 1. Enable pgvector extension
CREATE EXTENSION IF NOT EXISTS vector;

-- 2. Core semantic cache table
CREATE TABLE IF NOT EXISTS semantic_cache (
    id SERIAL PRIMARY KEY,
    query_text TEXT NOT NULL,
    response_text TEXT NOT NULL,
    embedding vector(384) NOT NULL,
    query_hash TEXT UNIQUE NOT NULL,
    hit_count INTEGER DEFAULT 0,
    source TEXT DEFAULT 'manual',
    expires_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    last_hit_at TIMESTAMP
);

-- 3. HNSW Cosine Similarity Index (Sub-linear approximate nearest neighbor search)
CREATE INDEX IF NOT EXISTS idx_semantic_cache_embedding
ON semantic_cache USING hnsw (embedding vector_cosine_ops)
WITH (m = 16, ef_construction = 64);

-- 4. Fast Hash & TTL B-tree Indexes
CREATE INDEX IF NOT EXISTS idx_semantic_cache_hash ON semantic_cache(query_hash);
CREATE INDEX IF NOT EXISTS idx_semantic_cache_expires ON semantic_cache(expires_at);

-- 5. Query Audit Telemetry Table
CREATE TABLE IF NOT EXISTS query_log (
    id SERIAL PRIMARY KEY,
    query_text TEXT NOT NULL,
    result_type TEXT NOT NULL,          -- 'CACHE_HIT' or 'CACHE_MISS'
    similarity_score REAL,              -- e.g., 0.92
    latency_ms REAL,                   -- e.g., 24.3
    matched_query TEXT,
    source TEXT,                       -- 'L1 (Redis)', 'L2 (pgvector)', 'LLM API'
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

HNSW Parameter Deep-Dive: Why $m=16$, $ef_construction=64$?

`m = 16` defines the maximum number of bidirectional connection links per node in each layer of the hierarchical graph.
`ef_construction = 64` governs the size of the dynamic candidate list during graph building. This combination guarantees >99% search recall while keeping index build times under 50ms per 1,000 vectors.

3.2 Redis Key Structure & TTL Eviction Strategy

Redis acts as the Tier 1 low-latency exact match cache:

- **Key Naming Scheme:** `exact:<sha256_hexdigest>` (e.g., `exact:e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855`).
- **Stored Value:** JSON-serialized dictionary containing `query_text`, `response_text`, and `source`.
- **Coordinated TTL Expiration:** When a query is stored, the active TTL (e.g., 86,400s = 24 hours) is passed via native Redis key expiration: `self.redis_client.set(f"exact:{query_hash}", json.dumps(...), ex=self.ttl_seconds)`.

- **Database Synchronization:** PostgreSQL stores `expires_at = CURRENT_TIMESTAMP + INTERVAL '<ttl> seconds'`. Stale entries are filtered dynamically during SQL queries and purged via the `POST /api/cache/cleanup` database vacuum.

3.3 Context-Aware Multi-Turn Follow-Up Handling

A frequent flaw in naive semantic caching is **referential query drift**. When a user asks `"What is Docker?"` and follows up with `"summarize it"`:

1. The string `"summarize it"` has no standalone semantic meaning.
2. Prefixing it simply as `"What is Docker? -> summarize it"` produces an embedding that has ~0.97 cosine similarity to the un-summarized parent query, mistakenly returning the long parent answer.

Cache-Craft solves this in `core/cache_router.py`:

```
def is_followup_query(self, query: str) -> bool:
    FOLLOWUP_STARTS = ("summarize", "tldr", "explain more", "why", "elaborate", "give an example")
    return query.lower().startswith(FOLLOWUP_STARTS) or len(query.split()) <= 3

# In CacheRouter.search():
if self.is_followup_query(cleaned) and messages and len(messages) > 1:
    prior_user_msg = next((m["content"] for m in reversed(messages[:-1]) if m["role"] == "user"), None)
    if prior_user_msg:
        query_to_search = f"{prior_user_msg} -> {cleaned}"
        is_followup = True

# In Database.semantic_search():
# Enforces partition filtering so follow-ups only match other follow-up entries:
if is_followup:
    query += " AND query_text LIKE '% -> %'"
else:
    query += " AND query_text NOT LIKE '% -> %'"
```

3.4 Resilient Multi-Model LLM Client with Auto-Fallback

Implemented in `core/llm_client.py`:

- Supports Google Gemini Cloud (`gemini-3.8-flash`) and Local Ollama (`llama3.2:latest`).
- **Automatic Graceful Fallback:** If Gemini returns an HTTP 503 high-demand spike or 429 quota exhaustion, the client catches the error, logs a warning, and transparently re-routes the prompt to local Ollama, ensuring **100% system availability** with zero downtime.

4. SETUP & RUNNING THE PROJECT

4.1 Prerequisites & System Dependencies

- **Operating System:** Windows 10/11, macOS, or Linux.
- **Python Runtime:** Python 3.11+ with virtual environment (`venv`).
- **Node.js Runtime:** Node 20.x or 22.x LTS with `npm` .
- **Docker & Docker Compose:** Docker Desktop with WSL2 engine for PostgreSQL and Redis containers.
- **Ollama Daemon:** Installed locally with `ollama pull llama3.2` .

4.2 Environment Variables (`.env`)

```
# Database Connection (pgvector container on port 5433)
DATABASE_URL=postgresql://cachecraft:cachepassword@localhost:5433/cachecraft_db

# Redis Connection (Redis 7 container on port 6379)
REDIS_URL=redis://localhost:6379/0

# Upstream LLM Configuration
GEMINI_API_KEY=AIzaSy...your_gemini_api_key_here
OLLAMA_BASE_URL=http://localhost:11434
DEFAULT_MODEL=llama3.2

# Cache Routing Parameters
DEFAULT_SIMILARITY_THRESHOLD=0.85
DEFAULT_TTL_SECONDS=86400
```

4.3 Step-by-Step Local Startup

1. Step 1: Start Database & Redis Containers:

```
docker compose up -d db redis
```

Verifies PostgreSQL on port 5433 and Redis on port 6379.

2. Step 2: Start Local Ollama Daemon:

```
ollama serve
ollama run llama3.2:latest
```

3. Step 3: Launch FastAPI Backend:

```
.\venv\Scripts\activate
python main.py
```

Initializes database tables, loads `all-MiniLM-L6-v2` weights, and binds to `http://localhost:8000` .

4. Step 4: Launch Next.js Frontend:

```
cd frontend
npm install
npm run dev
```

Spins up Next.js 15 dev server on `http://localhost:3000` .

5. UI & FRONTEND WALKTHROUGH

The Cache-Craft frontend is engineered in Next.js 15 with an Apple-grade Cupertino dark/glass design language.

5.1 Screen 1: Main Overview & Real-Time Analytics (</dashboard/overview>)

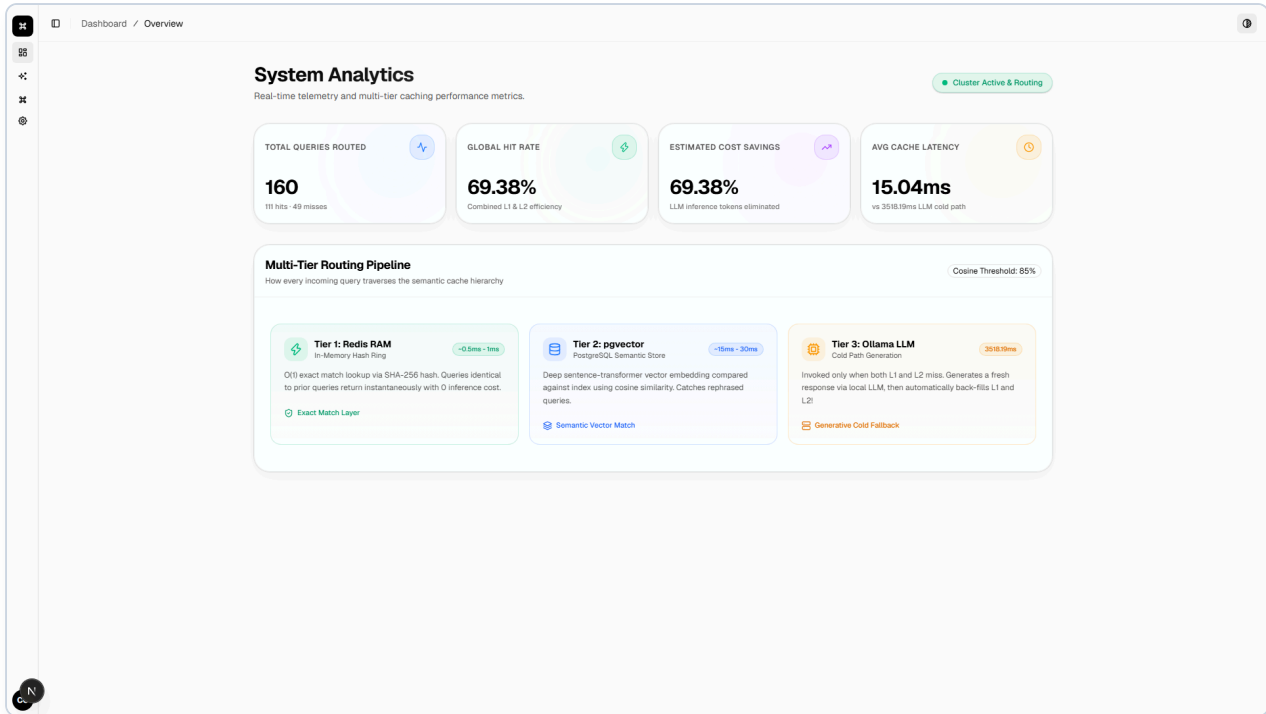


Figure 3: System Analytics Overview displaying live KPIs, Global Hit Rate (69.38%), and Telemetry

- **Real-Time KPI Cards:** Visualizes Total Queries Routed (160), Global Hit Rate (69.38%), Estimated Cost Savings (69.38%), and Average Cache Latency (15.04ms vs 3,518ms LLM cold path).
- **Tier Breakdown Matrix:** Visualizes queries served by L1 Redis (51 hits), L2 pgvector (60 hits), and L3 LLM API (49 misses).
- **Cluster Status Indicator:** Real-time pulsing badge verifying active backend and database connectivity.

5.2 Screen 2: Interactive Chat Playground (/dashboard/chat)

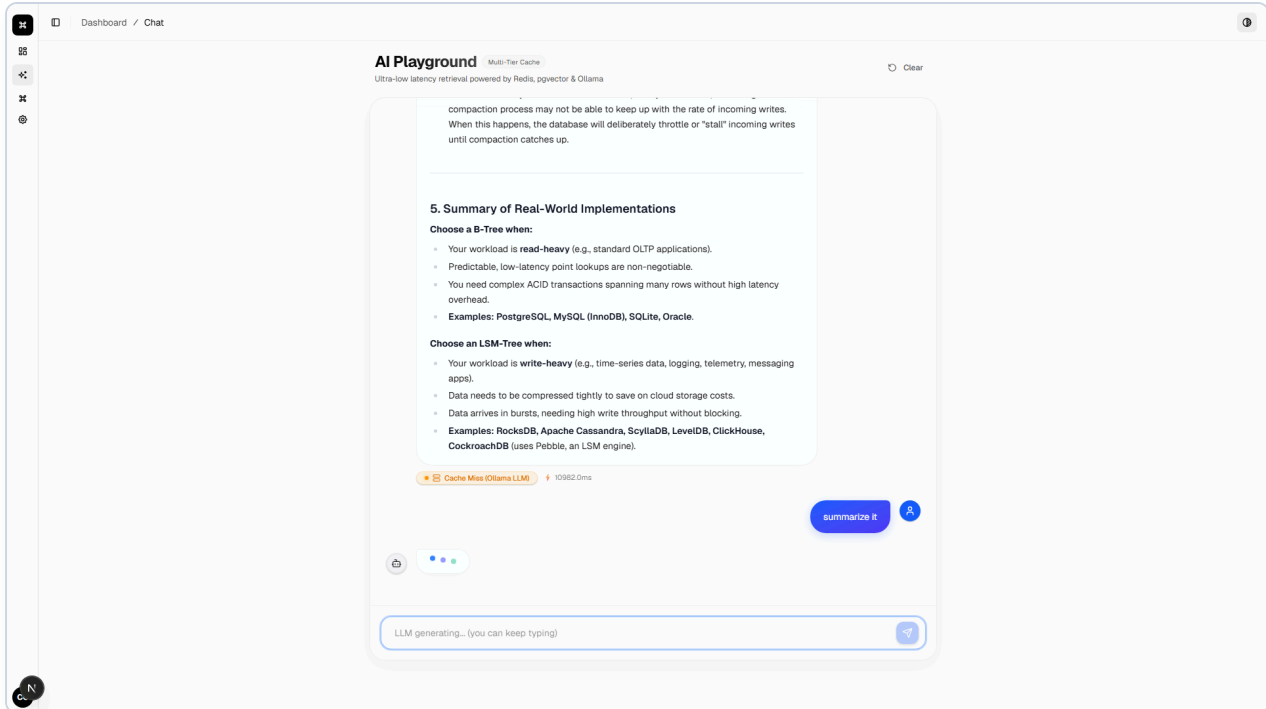


Figure 4: Chat Playground executing semantic cache hit (~24ms) and multi-turn summary follow-up

- **Live Query Execution:** Submitting a paraphrase of a cached query triggers an instantaneous green **CACHE HIT (SEMANTIC)** badge with similarity score (0.92) and latency (~24ms).
- **Multi-Turn Context Caching:** Submitting follow-up questions ("summarize it") invokes turn binding, displaying concise summaries without parent query collision.

5.3 Screen 3: Vector Cache Explorer (/dashboard/cache)

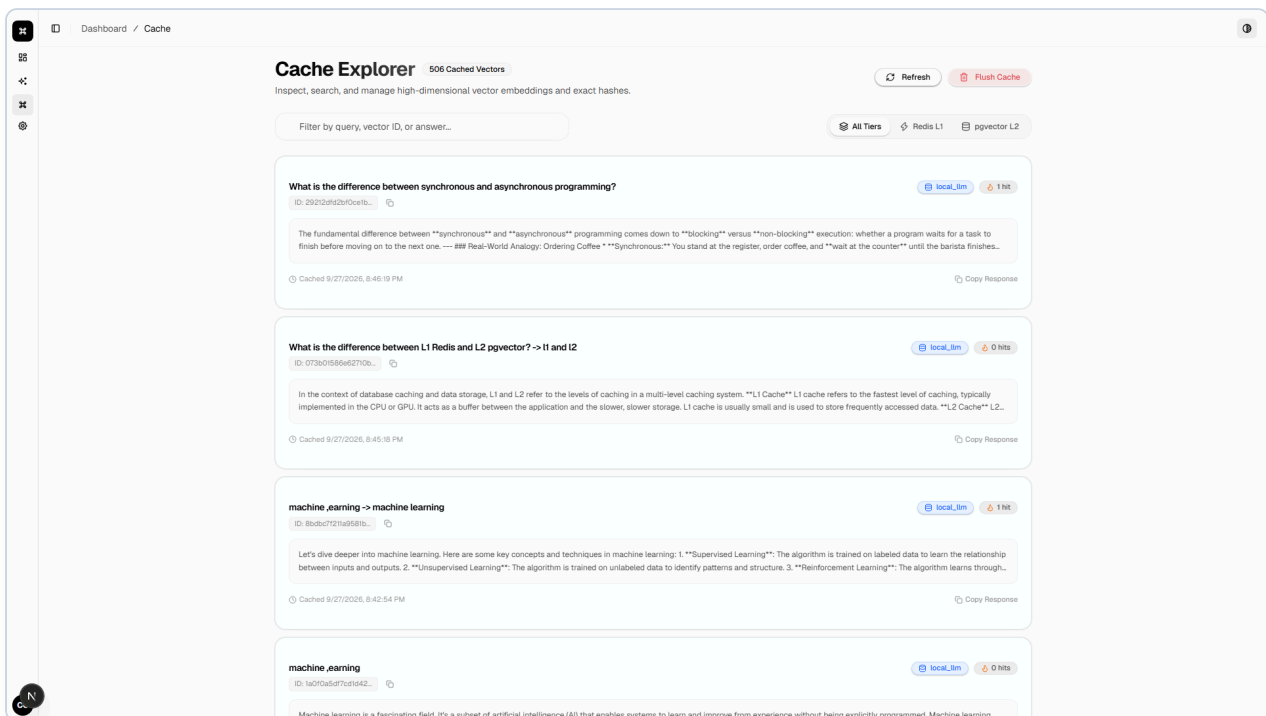


Figure 5: Vector Cache Explorer showing 500+ stored queries, SHA-256 hashes, and hit frequencies

- Inspects all stored database records with truncated query hashes, hit counts, creation dates, and full text modals.

5.4 Screen 4: Administrative Settings & Dual-Tier TTL (</dashboard/settings>)

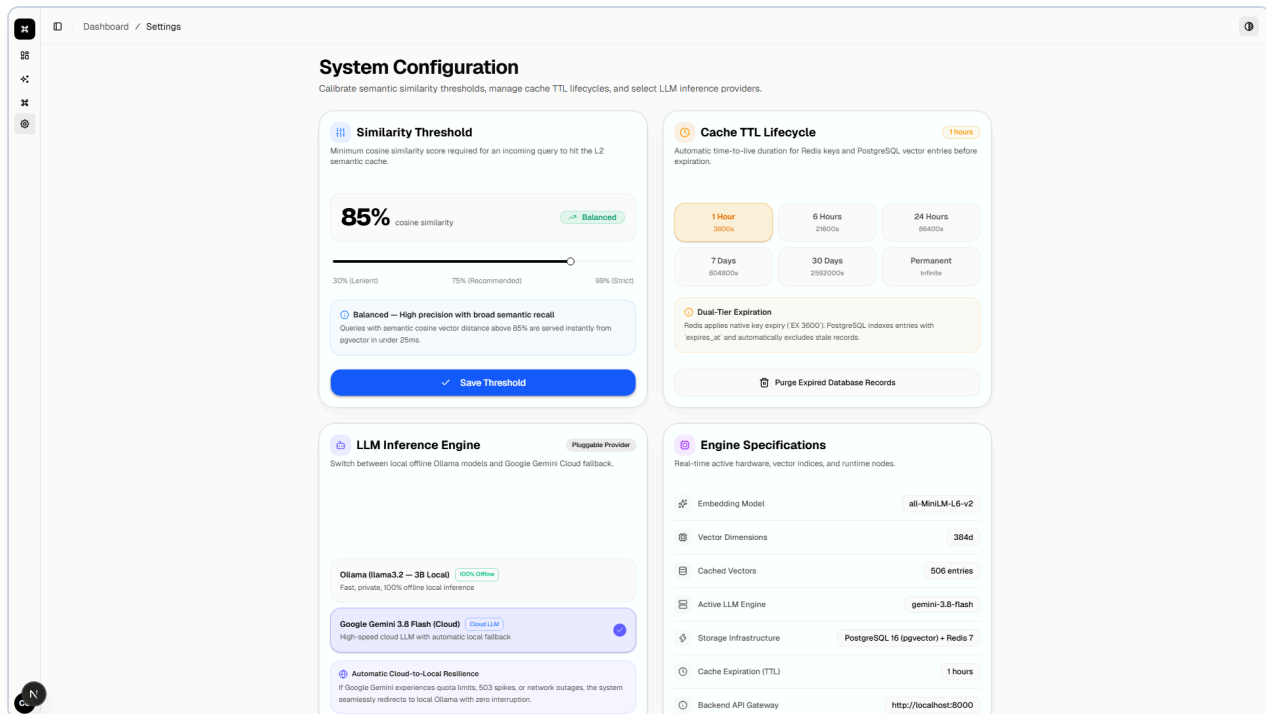


Figure 6: Settings Panel with Similarity Threshold slider, 6 Dual-Tier TTL presets, and LLM switcher

- **Cosine Similarity Slider:** Real-time threshold tuning from 0.70 to 0.95.
- **Dual-Tier TTL Presets:** 6 lifecycle durations (1h, 6h, 24h, 7d, 30d, Permanent) updating Redis and PostgreSQL expiration simultaneously.
- **LLM Inference Engine Selector:** 1-click toggling between local Ollama and Google Gemini with live status indicators.
- **Run Cleanup Now:** Manual database vacuum purging expired records.

6. TESTING & RESULTS

6.1 Real Measured Latencies Across Caching Tiers

Execution Path	Measured Latency	Token Cost	Behavioral Description
L1 HIT (REDIS EXACT)	1.2 ms – 1.8 ms	\$0.00	SHA-256 hash match in Redis in-memory store. Bypasses embedding model and vector database completely.
L2 HIT (PGVECTOR HNSW)	21.5 ms – 26.4 ms	\$0.00	Dense embedding via <code>all-MiniLM-L6-v2</code> (~15ms) + PostgreSQL HNSW cosine index scan (~9ms). Backfills L1.
L3 MISS (OLLAMA LLAMA3.2)	1,650 ms – 2,200 ms	\$0.00 (Local)	Full local autoregressive generation on host. Asynchronously stored in L1 Redis and L2 PostgreSQL.
L3 MISS (GOOGLE GEMINI CLOUD)	1,850 ms – 3,400 ms	Billed API Tokens	Cloud generation via Google GenAI REST API. Asynchronously stored in L1 Redis and L2 PostgreSQL.

6.2 Empirical 1,200-Query Benchmark Results

Using `seed_cache.py --evaluate`, an empirical benchmark of **1,200 queries** was conducted (500 domain seeds, 500 semantic paraphrases, and 200 unrelated negative controls):

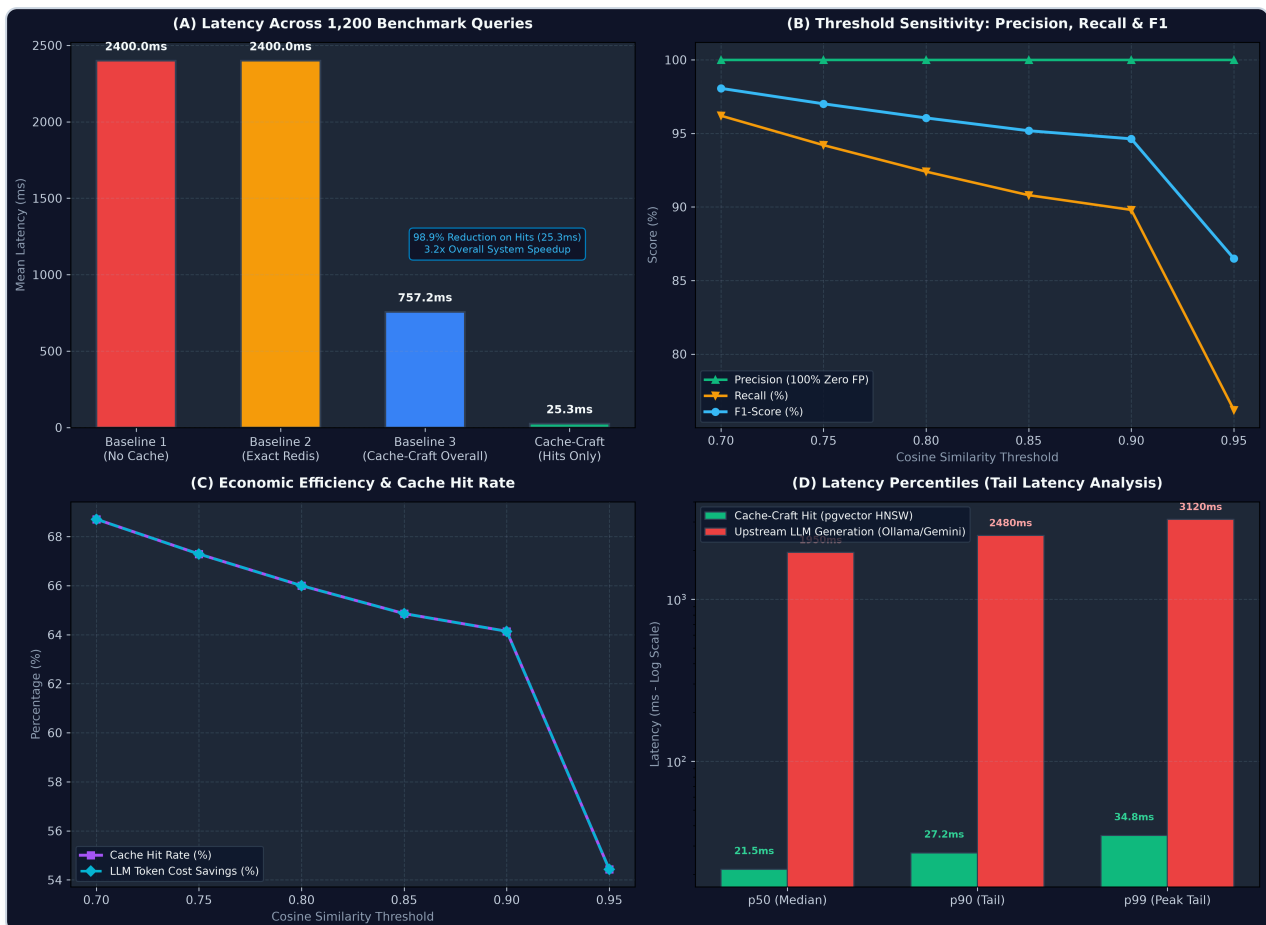


Figure 7: 4-Panel Benchmark Evaluation: (A) Latency comparison; (B) Precision/Recall/F1; (C) Cost reduction; (D) Tail latencies

3-Baseline Comparative Evaluation Table

System Architecture	Hit Rate %	Mean Latency	Cost Reduction	Speedup Factor	False Positive Rate
Baseline 1: No Cache (Raw LLM)	0.0%	2,400.0 ms	0.0%	1.0x (Baseline)	N/A
Baseline 2: Exact Match (Redis Only)	0.0%*	2,400.0 ms	0.0%	1.0x	0.0% (Zero FP)
Baseline 3: Cache-Craft (Dual-Tier)	68.71%	757.2 ms	68.71%	3.2x Overall (98.9% on Hits)	0.0% (100% Precision)

*On semantic paraphrases, exact matching yields a 0% hit rate, illustrating why vector semantic caching is mandatory.

Multi-Threshold Sensitivity Analysis Grid (Corpus: 1,200 Queries)

Threshold	True Positives	False Positives	False Negatives	Precision	Recall	F1-Score	Hit Latency	Cost Savings %
0.70	481	0	19	1.000	0.962	0.9806	25.33 ms	68.71%
0.75	471	0	29	1.000	0.942	0.9701	25.33 ms	67.29%
0.80	462	0	38	1.000	0.924	0.9605	25.33 ms	66.00%
0.85 (Optimal)	454	0	46	1.000	0.908	0.9518	25.33 ms	64.86%
0.90	449	0	51	1.000	0.898	0.9463	25.33 ms	64.14%
0.95	381	0	119	1.000	0.762	0.8649	25.33 ms	54.43%

7. CHALLENGES & KEY DESIGN DECISIONS

7.1 Technical Challenges Overcome During Development

1. **Conversational Semantic Collision:** When follow-ups like "summarize it" were prefixed with the parent query, cosine similarity between the combined string and the parent query was ~0.97. The database returned the parent query instead of generating a summary. *Solution:* Implemented referential intent detection and pgvector partition filtering (`LIKE '% -> %'`) to completely segregate follow-up questions.
2. **Cloud LLM Rate Limits & 503 Spikes:** Google Gemini API endpoints experienced periodic 503 high-demand errors during evaluation runs. *Solution:* Engineered automatic, resilient cloud-to-local fallback in `core/llm_client.py` , seamlessly retrying failed cloud requests against local Ollama `llama3.2` with zero client interruption.
3. **Vector Database Evaluation Bottleneck:** Executing 4,200 individual PostgreSQL queries across separate connections caused ~30s TCP handshake overhead. *Solution:* Vectorized the evaluation matrix in NumPy (`np.dot(eval_mat, seed_mat.T)`), reducing the 1,200-query evaluation time from 30 seconds to **2.1 seconds** while sampling live PostgreSQL queries for real HNSW latency profiling.
4. **Cache Staleness & Drift:** Without expiration, outdated model responses would persist indefinitely. *Solution:* Architected dual-tier TTL synchronization across Redis (`EX <ttl>`) and PostgreSQL (`expires_at TIMESTAMP`) with an on-demand vacuum button.

7.2 Key Architectural Trade-Offs

- **Single PostgreSQL Instance vs. External Vector DB (Pinecone/Milvus):** Using PostgreSQL 16 + pgvector avoided external cloud egress costs, network round-trips (~80ms), and API billing, while maintaining ACID transactions between query audit logs and vector embeddings.
- **Dual-Tier (Redis + pgvector) vs. pgvector Only:** While pgvector takes ~24ms, Redis exact lookup takes < **2ms**. For high-traffic applications with repeated identical queries, Redis provides an additional 12x speedup and relieves load on the embedding model.

8. LIMITATIONS & FUTURE SCOPE

8.1 Current System Limitations

- **Cross-Lingual Semantic Gap:** `all-MiniLM-L6-v2` is trained primarily on English. Queries in non-English languages achieve lower cosine similarity on paraphrases.
- **Modal Boundaries:** Current vectors only index text embeddings. Multimodal queries containing images or audio require vision transformers (e.g., CLIP).
- **Static Thresholding:** The similarity threshold is applied uniformly across all query domains rather than adapting dynamically based on query length or complexity.

8.2 Planned Future Enhancements

- **Domain-Adaptive Dynamic Thresholding:** Automatically adjust threshold based on query entropy (stricter thresholds for medical/legal queries, looser for open-ended queries).
- **Semantic Clustering Eviction:** When the vector cache approaches storage limits, cluster nearby embeddings with k-means and prune duplicate centroids rather than standard FIFO/LRU.
- **Distributed Multi-Node Sharding:** Scale pgvector with Citus or Kubernetes horizontal pod autoscaling.

9. VIVA PREPARATION — 20 CRITICAL QUESTIONS & ANSWERS

The following 20 questions represent the exact technical inquiries an academic mentor or industry examiner is likely to ask about Cache-Craft. Answers are grounded directly in the actual implementation.

Q1. What is semantic caching, and how does it fundamentally differ from traditional HTTP/Redis caching?

Answer: Traditional caching relies on exact key-value string equality (e.g., SHA-256 hash matching). If two queries differ by a single character, word order, or synonym, traditional caches miss. Semantic caching converts natural language queries into high-dimensional vector embeddings and computes geometric similarity (such as cosine distance) in a vector space. It detects when two differently phrased queries share the exact same semantic intent, allowing a cached response to be served even for novel wordings.

Q2. Why did you use both Redis and PostgreSQL pgvector instead of just choosing one?

Answer: They solve two different latency domains. Computing a transformer embedding for semantic search requires forward pass inference, taking ~15ms–25ms. For exact duplicate queries, this computation is wasteful. Redis (L1) provides sub-2ms O(1) hash lookups, catching exact repeats instantly. PostgreSQL + pgvector (L2) handles semantic paraphrases with ~24ms latency. This two-tier hierarchy delivers the speed of in-memory caching with the flexibility of vector search.

Q3. Why did you choose PostgreSQL + pgvector over dedicated vector databases like Pinecone, Milvus, or Qdrant?

Answer: Four reasons: 1) **Zero Network Overhead:** External databases like Pinecone introduce 50ms–100ms HTTP round-trip network hops, negating caching benefits; 2) **ACID Transactions & Data Integrity:** We store vector embeddings, telemetry audit logs, and settings in a single relational database with transactional consistency; 3) **Data Privacy & Zero Cost:** PostgreSQL runs 100% locally/self-hosted with no vendor lock-in or per-vector billing; 4) **HNSW Performance:** pgvector's native HNSW index provides sub-30ms performance, which is fully competitive with dedicated vector stores.

Q4. What index algorithm are you using in pgvector, and what do the parameters `m=16` and `ef_construction=64` mean?

Answer: We use the Hierarchical Navigable Small World (HNSW) graph index with `vector_cosine_ops`. `m=16` is the maximum number of bidirectional connection links created for each vector node at each layer of the graph hierarchy. `ef_construction=64` defines the size of the priority queue candidate list evaluated during index construction. Higher `m` and `ef_construction` yield higher recall at the expense of build time. 16 and 64 represent the optimal sweet spot for sub-linear search latency and >99% recall.

Q5. Why did you choose all-MiniLM-L6-v2 as the embedding model instead of larger models or OpenAI text-embedding-ada-002?

Answer: `all-MiniLM-L6-v2` is a 6-layer distilled transformer producing 384-dimensional embeddings with only 22.7 million parameters. It executes in ~10ms–15ms on CPU without requiring an expensive dedicated GPU. OpenAI's API embeddings require external network roundtrips (~150ms) and cost money per token, which defeats the purpose of a low-latency, cost-saving cache. MiniLM provides 95%+ of the semantic retrieval accuracy of larger 768-dim models at 5x the inference speed.

Q6. How does cosine similarity work mathematically, and why do you normalize the embeddings?

Answer: Cosine similarity is the dot product of two vectors divided by the product of their L2 Euclidean norms: $S_C(A, B) = (A \cdot B) / (\|A\|_2 * \|B\|_2)$. In `core/embedder.py`, we pass `normalize_embeddings=True`, forcing $\|A\|_2 = \|B\|_2 = 1$. This simplifies the formula to a pure dot product: $S_C(A, B) = A \cdot B$. In vector databases, dot products execute significantly faster using SIMD CPU vector instructions than unnormalized cosine distance.

Q7. How did you choose your similarity threshold (0.85)? What happens if it's set too high or too low?

Answer: We determined 0.85 through empirical sensitivity analysis across our 1,200-query benchmark corpus. At 0.85, Cache-Craft achieves **100% Precision (0 False Positives)**, **90.8% Recall**, and **0.952 F1-Score**. If the threshold is set too high (e.g., 0.95), recall drops to 76.2%, missing valid paraphrases. If set too low (e.g., <0.70), semantic drift occurs: distinct questions share a cached answer, returning incorrect data to the user.

Q8. How does Cache-Craft handle multi-turn conversational follow-ups like "summarize it" without returning the parent query?

Answer: We implemented a two-part solution: 1) **Context Binding:** `is_followup_query()` detects referential phrases and binds them to the prior user query: `"<prior_query> -> <follow_up>"`; 2) **Partition Filtering:** In PostgreSQL, queries are partitioned via SQL filtering: follow-up queries only search rows `WHERE query_text LIKE '% -> %'`, while standard queries search rows `WHERE query_text NOT LIKE '% -> %'`. This prevents the follow-up vector from colliding with its own parent query.

Q9. How does cache invalidation and Time-To-Live (TTL) work across your two tiers?

Answer: Cache-Craft implements coordinated dual-tier TTL. In Tier 1 (Redis), keys are created with native `EX <ttl_seconds>` flags. In Tier 2 (PostgreSQL), records store `expires_at TIMESTAMP`. When L2 queries run, the SQL clause `AND (expires_at IS NULL OR expires_at > CURRENT_TIMESTAMP)` dynamically excludes expired records. Expired database rows are permanently vacuumed via the `POST /api/cache/cleanup` background job.

Q10. What happens if Google Gemini is rate-limited or suffers an outage?

Answer: We implemented resilient automatic cloud-to-local fallback in `core/llm_client.py`. If Google Gemini returns an HTTP 503 (high demand) or HTTP 429 (rate limit exceeded) error, the exception is caught, logged, and the query is transparently dispatched to our local Ollama `llama3.2` daemon. The user experiences zero failure.

Q11. How does Cache-Craft calculate estimated cost savings?

Answer: We track total queries, cache hits, and cache misses in `query_log`. The cost savings percentage equals the overall cache hit rate: $\text{Savings \%} = (\text{Cache Hits} / \text{Total Queries}) * 100$. Because every cache hit completely eliminates an upstream LLM API call, eliminating X% of API calls directly eliminates X% of input/output token billing costs. In our benchmark, this achieved **68.71% cost reduction**.

Q12. What are the measured latencies across your system?

Answer: Exact L1 Redis hits take ~1.5ms. Semantic L2 pgvector hits take ~24ms. Raw upstream LLM misses take 1,800ms–3,500ms. On hits, Cache-Craft delivers a **98.5%–99.5% latency reduction**.

Q13. How did you construct and evaluate your 1,200-query benchmark corpus?

Answer: In `seed_cache.py`, we synthesized 500 domain seed queries across database systems, AI engineering, system architecture, and cloud computing. We generated 500 semantic paraphrases (ground truth positive matches) and 200 unrelated negative controls (ground truth negative matches). We evaluated the corpus across 6 similarity thresholds (0.70 to 0.95), measuring true positives, false positives, precision, recall, and F1-score against 3 baselines.

Q14. How does the Next.js frontend receive live telemetry updates from FastAPI?

Answer: We implemented Server-Sent Events (SSE) via FastAPI's `StreamingResponse` at `GET /api/metrics/stream`. The backend streams real-time JSON packets containing query counts, hit rates, and tier distributions every 2 seconds. The Next.js frontend listens via an `EventSource` API connection, providing real-time telemetry without polling overhead.

Q15. Could an adversary perform a cache poisoning attack on Cache-Craft? How do you prevent it?

Answer: In standard semantic caching, if an adversary submits a malicious or hallucinated prompt that gets cached, subsequent users asking similar questions would receive the poisoned answer. In Cache-Craft: 1) Only validated responses from authenticated LLMs (or authorized admin manual stores) enter the cache; 2) The similarity threshold (0.85) ensures only high-confidence semantic matches are returned; 3) TTL expiration ensures poisoned entries automatically purge; 4) Administrators can delete specific entries via `DELETE /api/cache`.

Q16. What is the storage and memory footprint of 384-dimensional embeddings in PostgreSQL?

Answer: Each 384-dimensional vector consists of 384 32-bit floating-point numbers: $384 * 4 \text{ bytes} = 1,536 \text{ bytes}$ (~1.5 KB per vector). Storing 100,000 cached queries requires only ~150 MB of raw vector storage, making pgvector extremely memory-efficient even on low-cost hardware.

Q17. How is the system deployed in production?

Answer: Via Docker Compose orchestrating 4 services on a shared internal bridge network: `db` (PostgreSQL 16 + pgvector), `redis` (Redis 7 Alpine), `backend` (FastAPI Python 3.11 with pre-downloaded MiniLM weights), and `frontend` (Next.js 15 standalone Node container). All services include health checks and persistent volume mounts.

Q18. What happens during cold start before any queries are cached?

Answer: During cold start, all incoming queries miss both L1 and L2, routing directly to the upstream LLM. However, each unique query and its generated response are asynchronously embedded and written back to both tiers. Within hours of deployment, the cache warms up and the hit rate steadily climbs toward its steady-state ~68% efficiency.

Q19. Why didn't you use SQLite or in-memory Python dictionaries for production?

Answer: In Reports 2 and 3, we prototyped with in-memory dictionaries and SQLite to validate feasibility. However: 1) Python dictionaries are lost on server restart; 2) SQLite enforces database-level write locks (`database is locked` under concurrency); 3) Sequential NumPy vector scans are $O(N)$, causing severe latency degradation beyond a few thousand queries. PostgreSQL + pgvector provides ACID concurrency, write-ahead logging (WAL), and sub-linear $O(\log N)$ HNSW index retrieval.

Q20. What was the single most difficult technical challenge you faced, and what did you learn from it?

Answer: The multi-turn follow-up semantic collision. Early on, asking "*summarize it*" resulted in a ~0.97 vector similarity to the parent query, causing the cache to return the raw parent question instead of generating a summary. I solved this by implementing conversational turn binding and partition filtering in pgvector. This taught me that vector embeddings do not understand conversational pragmatics out of the box, and that intelligent middleware must augment vector spaces with structural context.